Purine/ Pyrimidine Metabolism

BIOSYNTHESIS OF PURINE NUCLEOTIDES

i. The purine nucleotides are synthesized by most of the tissues. However the major site is the liver. This pathway operates in the cytoplasm.

ii. The major pathway is denoted as ***de novo* synthesis**, because the purine ring is synthesized from different small components. Since the human being can synthesize the purine and pyrimidine bases *de novo*, they are said to be prototrophs.

iii. The contribution of different atoms from different sources for the formation of the purine ring is shown in the Figure 39.10.

iv. During *de novo* synthesis, *purine ring is built up on a ribose-5-phosphate molecule*. Hence nucleotides are the products of the *de novo* synthesis.

v. There are ten steps in the *de novo* synthesis pathway. The enzymes catalyzing these reactions are existing as a multienzyme complex in eukaryotic cells; this arrangement increases the efficiency of the pathway. The names of the enzymes catalyzing the purine synthesis steps are as follows (see Figs 39.12 and 39.13)

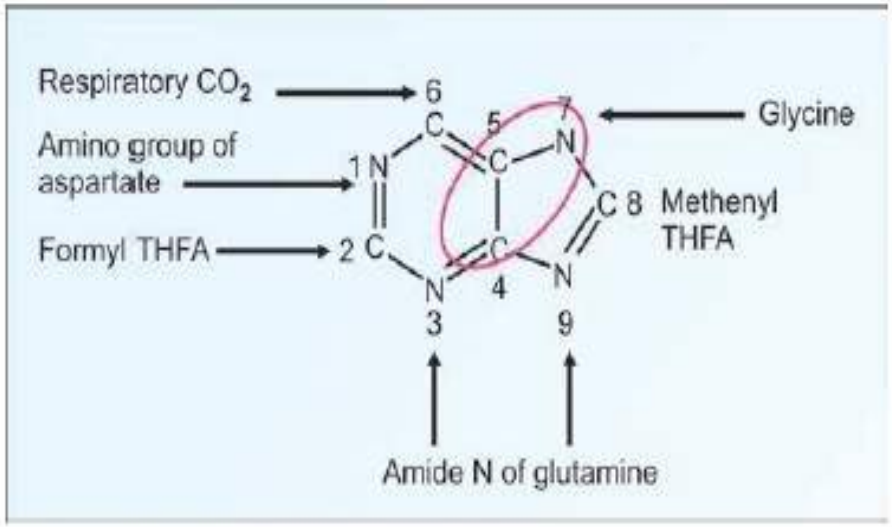


Fig. 39.10. The assembly of purine ring is from various sources. THFA (FH_4) = tetra hydro folic acid

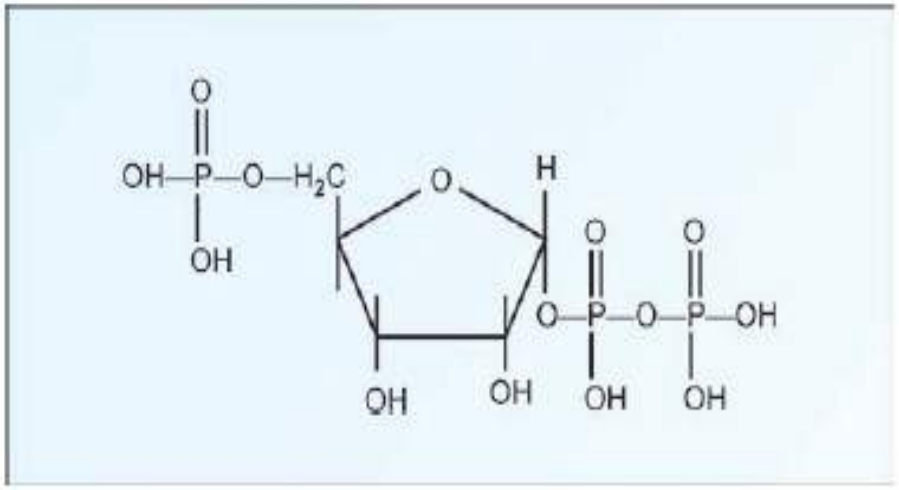


Fig. 39.11. Phospho ribosyl pyrophosphate

Step 0 (Preparatory Step), PRPP synthesis

i. Phospho ribosyl pyrophosphate (PRPP) is the donor of ribose-5-phosphate for *de novo* synthesis. The reaction is:
Ribose-5-phosphate + ATP →
ADP + Phospho ribosyl pyrophosphate (PRPP)
The structure of PRPP is shown in Figure 39.11.

ii. The purine ring is later on assembled on the ribose-5-phosphate.

iii. PRPP is also used for the synthesis of pyrimidine nucleotides, nucleotide co-enzymes and also for the salvage pathway. Hence the synthesis of PRPP is not considered as a step in the *de novo* synthesis of purine nucleotides; it is called a preliminary or preparatory step.

Formation of AMP

Steps 1 to 10 are summarized in Table 39.5. Flow diagrams of these steps are shown in Figures 39.12 and 39.13. The *de novo* synthesis of one molecule of purine nucleotide requires 6 ATP.

First step is the rate limiting enzyme. Step 4 is inhibited by **azaserine**, an anticancer drug. **6-mercapto-purine** inhibits amination of IMP to AMP, and so it is an **anticancer drug**.

Table 39.5. Summary of steps of purine synthesis

Step	Donor	Added atom	Product
1.	Glutamine	N9 (Rate limiting)	PRA
2.	Glycine (ATPrequired)	C4, 5, N7	GAR
3.	Methenyl-THFA	C8	FGAR
4.	Glutamine	N3 (ATP required)	FGAM
5.	–	Ring closure (ATP)	AIR
6.	Carbon dioxide	C6	ACAIR
7.	Aspartic acid	N1 (ATP required)	SAICAR
8.	–	Fumarate removed	AICAR
9.	Formyl-THFA	C2	FAICAR
10.	–	Ring closure	IMP

The expansions of the abbreviated forms of the products are shown in Figures 39.12 and 39.13.

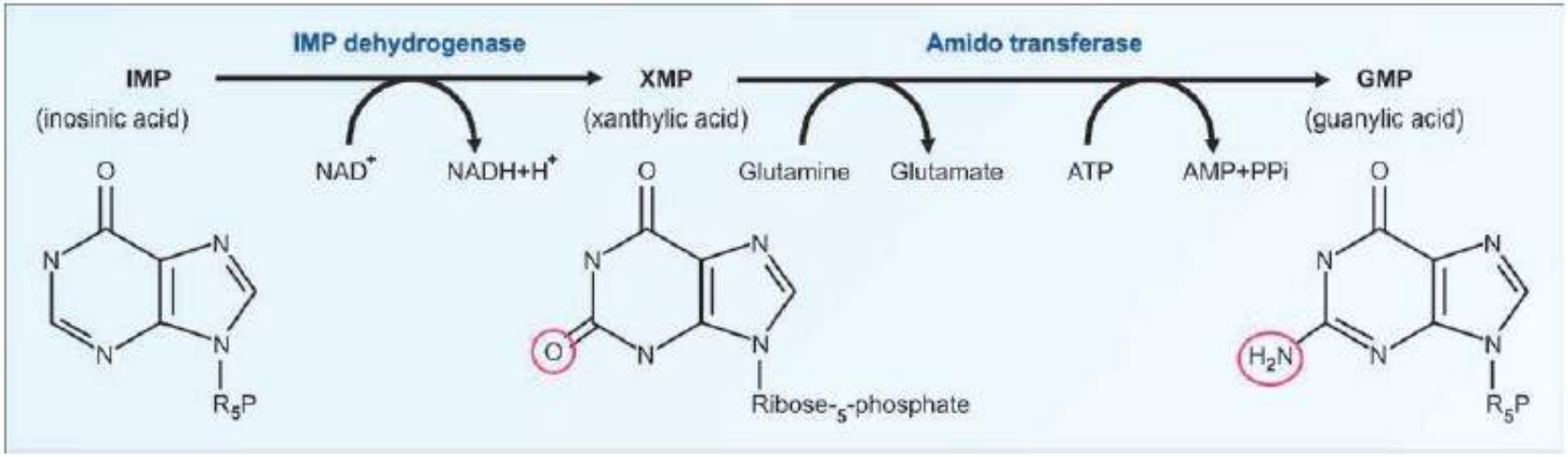


Fig. 39.14. Conversion of IMP to GMP. R-5-P = ribose-5-phosphate

Conversion of IMP to GMP

The conversion of IMP to GMP involves two steps, first oxidation of IMP to xanthylic acid (xanthosine monophosphate) (XMP) by an NAD^+ dependent dehydrogenase. Then an amido transferase transfers the NH_2 group from glutamine to XMP to form GMP. ATP is hydrolyzed to AMP level in this reaction (Fig. 39.14). Both AMP and GMP can be converted to their di- and tri- phosphates. *Synthesis of one molecule of purine nucleotide requires 6 high energy phosphates.*

Salvage Pathway

i. This pathway ensures the recycling of purines formed by degradation of nucleotides. Nucleosides and deoxy-nucleosides can also be salvaged.
ii. PRPP is the starting material in this pathway; it is also a substrate for *de novo* synthesis. Hence these two pathways are closely inter-related.

iii. The free purines are salvaged by two different enzymes; adenine phospho ribosyl transferase (**APRTase**) and hypoxanthine guanine phosphoribosyl transferase (**HGPRTase**).

iv. The pathway is of special importance in tissues like RBCs and brain where the *de novo* pathway is not operating. The salvage pathway economizes intracellular energy expenditure.

v. Absence of enzymes of salvage pathway produces specific clinical syndromes. Salvage pathway is summarized below:

$$\text{Adenine} + \text{PRPP} \xrightarrow{\text{APRTase}} \text{AMP} + \text{PPi}$$

$$\text{Guanine} + \text{PRPP} \xrightarrow{\text{HGPRTase}} \text{GMP} + \text{PPi}$$

$$\text{Hypoxanthine} + \text{PRPP} \xrightarrow{\text{HGPRTase}} \text{IMP} + \text{PPi}$$

Regulation of Purine Synthesis

i. The committed step in *de novo* synthesis is the reaction catalyzed by amido-transferase (step 1). It is inhibited by AMP and GMP.
ii. They act as allosteric modifiers. Binding of AMP and GMP on the enzyme converts monomeric active form to a dimeric inactive form.
iii. Since both AMP and GMP can bind to the same enzyme molecule at different sites, they act synergistically.
iv. Both AMP and GMP inhibit their own formation by feedback inhibition of adenylosuccinate synthetase and IMP dehydrogenase.
v. The formation of AMP from IMP requires GTP; similarly formation of GMP requires ATP. Hence both GTP and ATP are made available in sufficient quantities.
vi. The availability of PRPP is another important regulatory factor. The activity of PRPP synthetase is regulated by negative modifiers; purine and pyrimidine nucleotides.

Analogues as Purine Synthesis Inhibitors

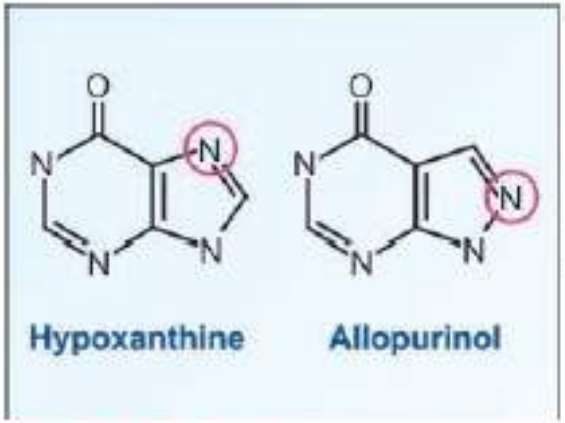
They act as **competitive inhibitors** of the naturally occurring nucleotides that are used to synthesize DNA. When wrong bases are incorporated, the DNA becomes functionally inactive. Thereby cell division is arrested. So they are useful as anticancer drugs. A few examples are:

a. Mercaptopurine inhibits the conversion of IMP to GMP and AMP.

b. Folate antagonists (Methotrexate) would affect the reactions involving one carbon group transfers.

c. Azaserine (diazo acetyl-L-Serine) is a glutamine antagonist and therefore inhibits reactions involving glutamine (steps 1 and 4).

d. Other synthetic nucleotide analogues used as anticancer agents are 6-thio guanine and 8-aza guanine.



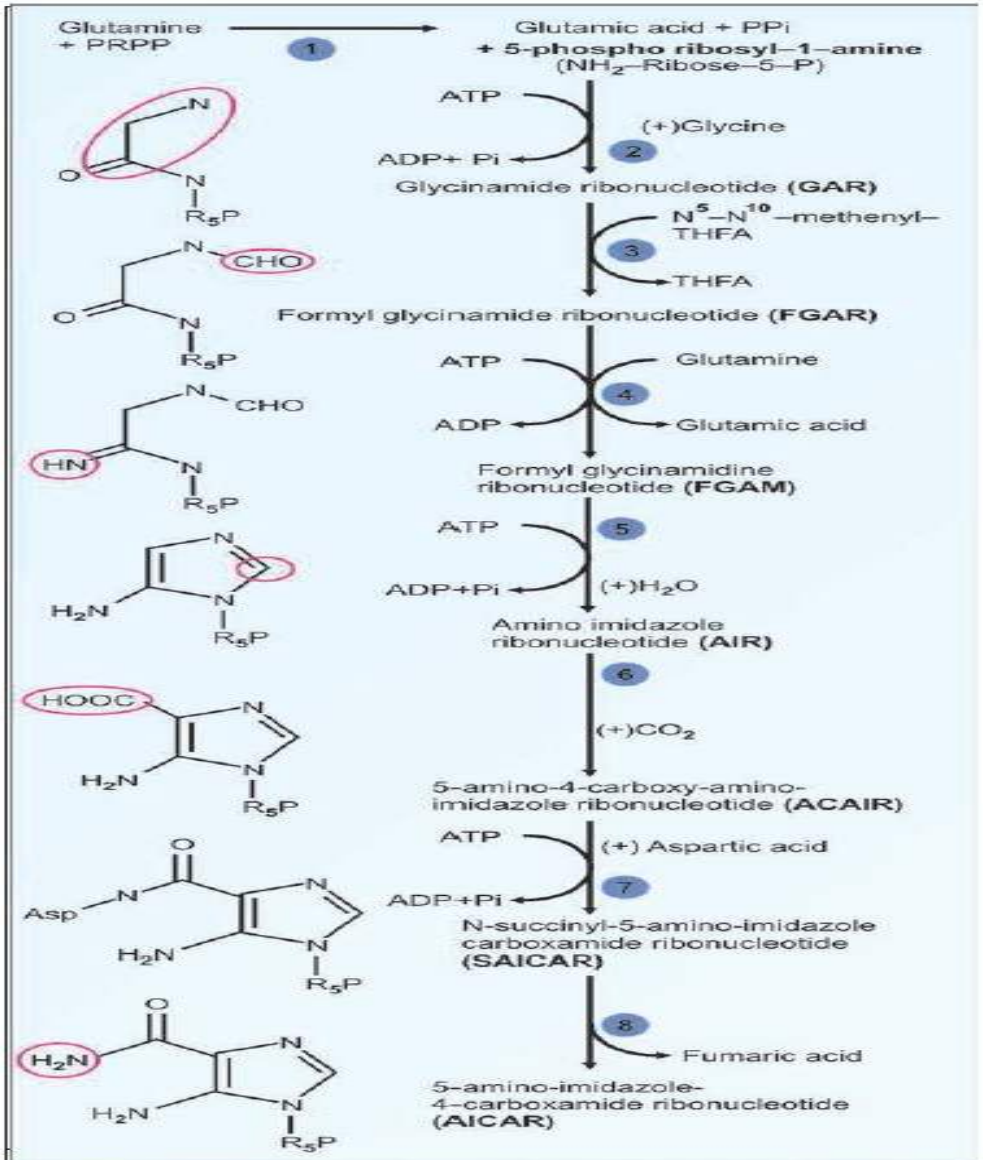


Gertrude
Elion
NP 1988
1918-1999



George
Hitchings
NP 1988
1905-1998

Fig. 39.16. Allopurinol inhibits xanthine oxidase; an example of competitive inhibition



PRPP= phosphoribosyl pyrophosphate. R-5-P = ribose-5-phosphate. Note that purine ring is built upon a ribose-5-phosphate. Names of enzymes of each step are shown in the text.

Fig. 39.12. First eight steps of purine synthesis

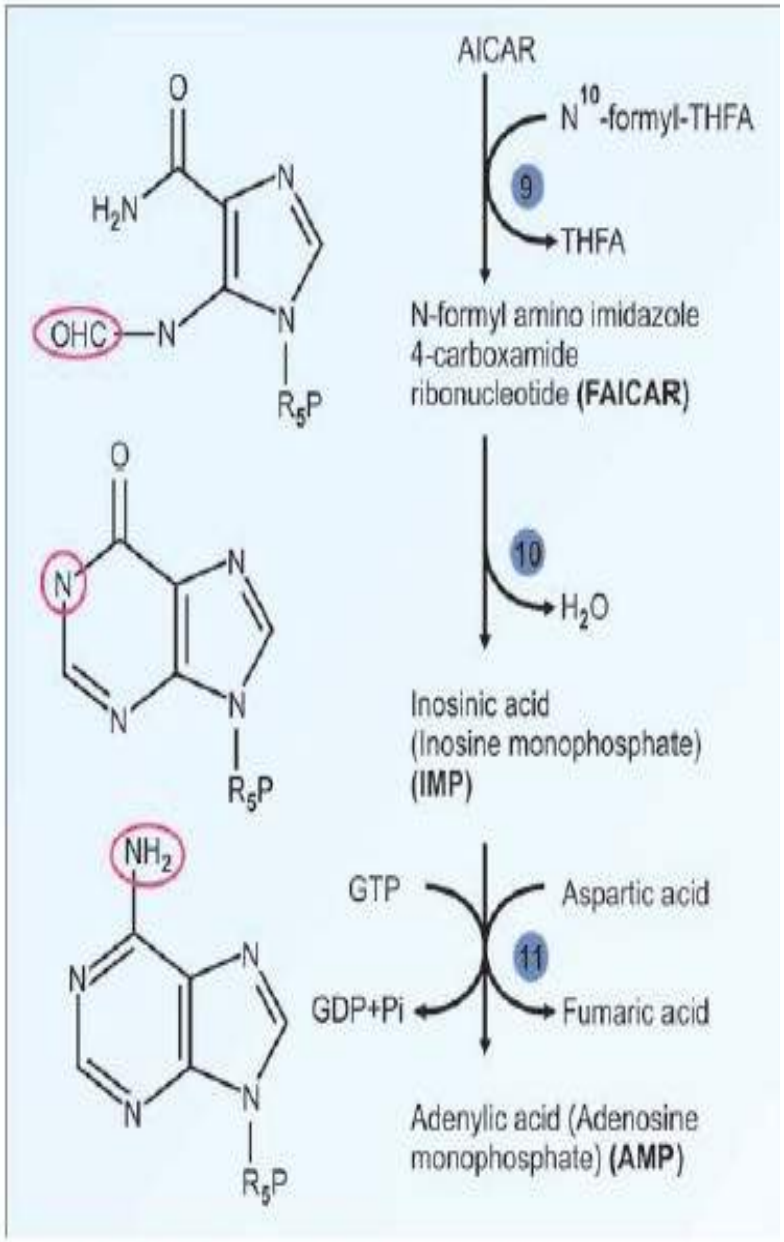


Fig. 39.13. Last steps in purine synthesis

Purine Nucleoside Di- and Tri-phosphates

These reactions are catalyzed by specific kinases and phosphate group is donated by ATP. One example is given below.

$$\text{GMP} + \text{ATP} \xrightarrow{\text{(Monophosphate kinase)}} \text{GDP} + \text{ADP}$$

$$\text{GDP} + \text{ATP} \xrightarrow{\text{(Diphosphate kinase)}} \text{GTP} + \text{ADP}$$

Degradation of Purine Nucleotides

The end product of purine nucleotide catabolism is **uric acid** (urate). The structure is shown in Figure 39.2. This degradation is taking place mainly in the liver. The steps are shown in Figure 39.15.

The xanthine oxidase is a metalloflavoprotein containing FAD, **molybdenum** and iron. As xanthine is oxidized to uric acid, the electrons are transferred first to molybdenum, then to FAD, and finally to molecular oxygen, when **hydrogen peroxide** (one of the reactive oxygen species) is produced.

Species difference

The end product of purine catabolism in human beings is uric acid. However, the total amount of nitrogen excreted as uric acid is very little, because human beings are **ureotelic**. The amino nitrogen is finally excreted as urea in mammals. The birds, amphibians and reptiles are **uricotelic** because they excrete uric acid as the major end product of purine as well as amino acid catabolism. The lower primates and some other mammals have the enzyme uricase which converts uric acid to allantoin and the final product excreted is allantoin which is more soluble.

Caffeine

Caffeine is the trimethyl derivative of xanthine. Coffee and tea contains caffeine. It inhibits phosphodiesterase, causes prolonged action of cyclic AMP and increases the activity of hormone sensitive lipase (Fig. 11.16). Caffeine enhances the effect of epinephrine on glycogenolysis.

URIC ACID

Strecker in 1857 showed the presence of uric acid in urine. In 1892, Sir Frederick Hopkins (Nobel prize, 1929) estimated uric acid. In 1895, Emil Fischer (Nobel prize, 1902) showed that uric acid is derived from purine nucleus.

Normal blood level of uric acid ranges from 2-5 mg/dl in females and 3-7 mg/dl in males. The daily excretion varies from 500-700 mg. Nucleic acid content is more in non-vegetarian diet. Uric acid is sparingly soluble in water.

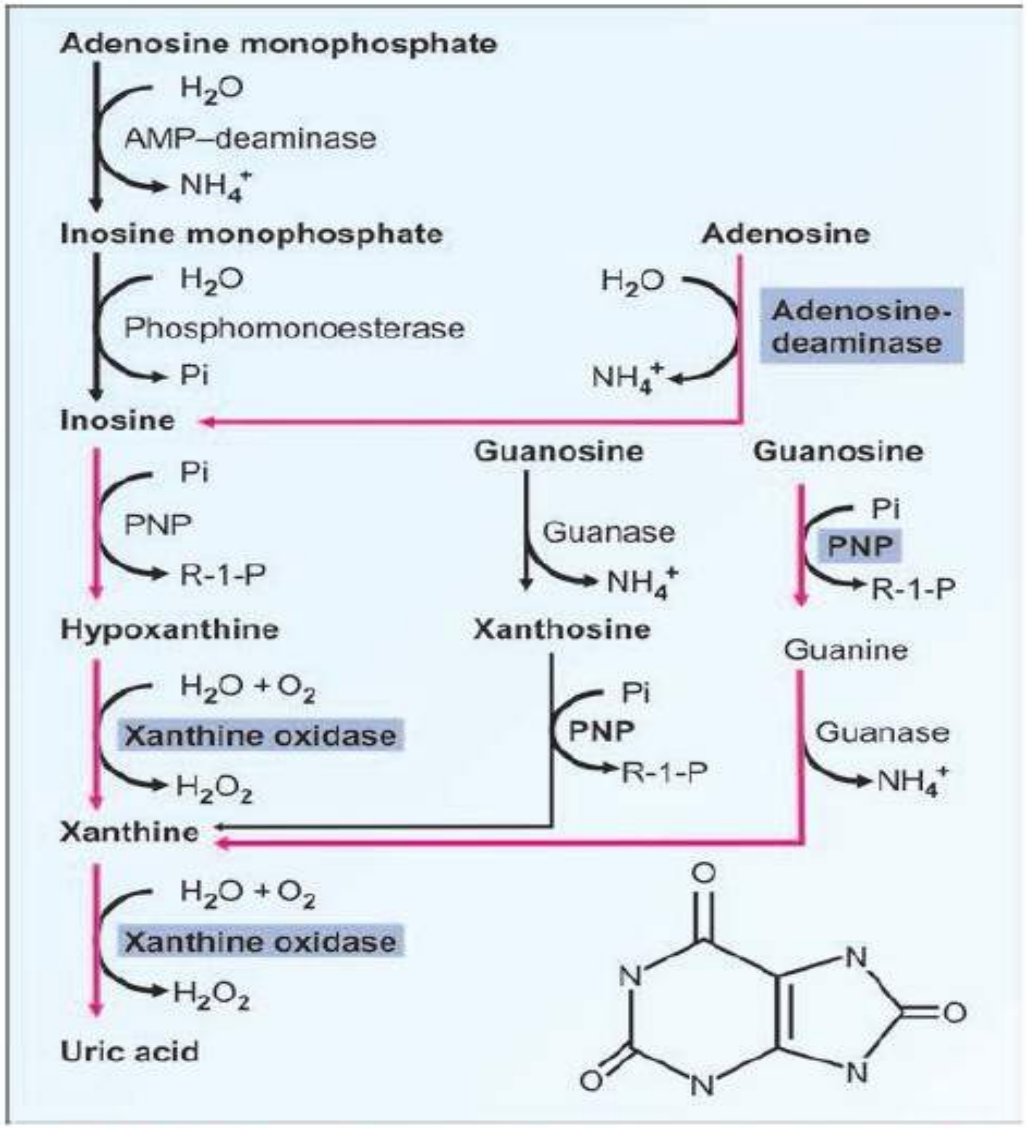


Fig. 39.15. Degradation of purine nucleotides. Main pathway is in red arrows. PNP = purine nucleoside phosphorylase; R-1-P = ribose-1-phosphate

Disorders of Purine Metabolism

The most common abnormality is an elevation of uric acid level in blood, referred to as **hyper-uricemia**. It is defined as serum uric acid concentration exceeding 7 mg/dl in male and 6 mg/dl in female. It may or may not be associated with increased excretion of uric acid in urine, which condition is called **uricosuria**. The manifestations are due to the low solubility of uric acid in water.

GOUT

i. It is due to accumulation of urate crystals in the synovial fluid resulting in inflammation leading to acute arthritis.
ii. At 30°C, the solubility of uric acid is lowered to 4.5 mg/dl. Therefore uric acid is deposited in cooler areas of the body to cause **tophi**. Thus tophi are seen in distal joints of foot.
iii. Increased excretion of uric acid may cause deposition of uric acid crystals in the urinary tract leading to **calculi** or stone formation with renal damage. Gout may be either primary or secondary

1. Primary Gout

About 10% of cases of primary gout are idiopathic. Primary gout may show a familial incidence. Incidence of primary gout is about 1:500 in total population. Causes of primary gout are:

1-A. 5-phosphoribosyl amido transferase

The abnormal enzyme is active, but not sensitive to feedback regulation by the inhibitory nucleotides. This would lead to overproduction of purine nucleotides.

1-B. Abnormal PRPP synthetase

The enzyme is not subject to normal allosteric control mechanisms and this could lead to increased production of PRPP. The condition is X-linked and recessive.

1-C. Deficiency of enzymes of salvage pathway

This would result in increased availability of PRPP and decreased purines; so feedback inhibition is lost.

1-D. Glucose-6-phosphatase deficiency

This condition is known as von Gierke's disease (glycogen storage disease, type I, see Chapter 9). When this enzyme is deficient, glucose-6-phosphate cannot be converted to glucose. So more glucose is channelled into the pentose-phosphate shunt pathway, resulting in increased availability of ribose-5-phosphate. This would lead to increased formation of PRPP.

1-E. Glutathione reductase variant

This enzyme depends on NADPH generated by pentose phosphate pathway. The abnormality in enzyme leads to increased production of ribose-5-phosphate and thereby increased level of PRPP. Dysregulation of the rate limiting step of purine nucleotide synthesis leads to increased synthesis and degradation of uric acid.

2. Secondary Hyperuricemia

2-A. Increased production of uric acid

It may be due to enhanced turnover rate of nucleic acids as seen in

i. Rapidly growing malignant tissues, e.g. leukemias, lymphomas, polycythemia.
ii. Increased tissue breakdown after treatment of large malignant tumors.
iii. Increased tissue damage due to trauma and raised rate of catabolism as in starvation.

2-B. Reduced excretion rate

i. Renal failure.
ii. Treatment with thiazide diuretics which inhibit tubular secretion of uric acid.
iii. Lactic acidosis and keto-acidosis due to interference with tubular secretion.

3. Clinical Findings of Gout

Galen has described the classical image of the gouty subjects as the red faced, good living, hard drinking, country squire. Many geniuses, including Isaac Newton, Gibbon and Johnson were suffering from gout.

Gouty attacks may be precipitated by high purine diet and increased intake of **alcohol**. Often the patients have a few drinks, go to sleep symptomless, but are awakened during the early hours of morning by excruciating joint pains. Alcohol leads to accumulation of lactic acid.

The typical gouty **arthritis** affects the first metatarsophalangeal joint (big toe), but other joints may also be affected. The joints are extremely painful. Synovial fluid will show birefringent urate crystals.

In **chronic cases**, uric acid may get deposited around joints causing swelling (**tophi**) composed of sodium urate. The total urate pool (normal 1200 mg) is increased to 3000 mg in gout patients without tophi. It may be as high as 30,000 mg in patients with tophi.

In chronic gout, the deposition of urate crystals in renal medulla occurs which progresses to urolithiasis and renal damage.

Treatment Policies in Gout

i. Reduce dietary purine intake and restrict alcohol.

ii. Increase renal excretion of urate by **uricosuric drugs**, which decrease the reabsorption of uric acid from kidney tubules, e.g. probenecid.

iii. Reduce urate production by **allopurinol**, an analogue of hypoxanthine (Fig. 39.16). Allopurinol is a **competitive** inhibitor of xanthine oxidase thereby decreasing the formation of uric acid. Xanthine and hypo-xanthine are more soluble and so are excreted more easily. Xanthine oxidase converts allopurinol to alloxanthine. It is a more effective inhibitor of xanthine oxidase. This is a good example of '**suicide inhibition**' (Chapter 5). Allopurinol was synthe-sised by Elion and Hitchings independently (Nobel prize, 1988).

iv. **Colchicine**, an anti-inflammatory agent is very useful to arrest the arthritis in gout.

Lesch-Nyhan Syndrome

It is an **X-linked** inherited disorder of purine metabolism. Incidence is 1:10,000 males. There is deficiency of **HGPRTase**. So, the rate of salvage pathway is decreased resulting in accumulation of PRPP and decreased level of inhibitory purine nucleotides. The disease is characterized by **self mutilation**, mental retardation, excessive uric acid and nephro-lithiasis. Gout develops in later life. The neurological manifestations suggest that the brain is dependent on the salvage pathway for the requirements of IMP and GMP.

Hypouricemia

Adenosine Deaminase (ADA) Deficiency

It is associated with severe immunodeficiency where both T and B cells are deficient. It is an inherited autosomal recessive disease. ADA deficiency leads to accumulation of adenosine and dATP; this would inhibit further production of precursors for DNA synthesis especially dCTP. Lymphocytes usually contain high quantity of ADA. Therefore ADA deficiency is mainly manifested as reduced lymphocytes. This leads to impaired cellular and humoral immunity. Hypouricemia is due to defective breakdown of purine nucleotides. Antibiotics and periodic injections of immunoglobulin will be life saving. Weekly intramuscular injections of bovine ADA were found to be beneficial. Bone marrow stem cells will increase both T and B cells in the patients. The first successful gene replacement therapy has been tried in ADA deficiency (Chapter 43).

Xanthine Oxidase Deficiency

It is a genetic defect. Characteristic features are hypouricemia, increased excretion of hypoxanthine and xanthine and liver damage.

DE NOVO SYNTHESIS OF PYRIMIDINE

The pyrimidine ring (unlike the purine) is synthesized as free pyrimidine and then it is incorporated into the nucleotide. The origin of atoms of pyrimidine nucleus is indicated in Figure 39.17.

Step 1: Carbamoyl Phosphate Synthesis

The reaction occurs in cytoplasm (in urea synthesis, the reaction is in mitochondria). The nitrogen of glutamine and bicarbonate react to form carbamoyl phosphate (step 1, Fig. 39.18). The enzyme is carbamoyl phosphate synthetase II (CPS II). (The differences between CPS-I and CPS-II are described in Table 14.2).

Step 2: Rate Limiting Step

Carbamoyl phosphate and aspartate combine to form **carbamoyl aspartate** (step 2, Fig. 39.18). The enzyme is aspartyl transcarbamoylase (ATC), which is allosterically regulated. The atoms C2 and N3 are derived from carbamoyl phosphate and the rest are from aspartate.

Step 3: Formation of Pyrimidine Ring

The 3rd nitrogen and 4th carbon are joined by a covalent bond and carbamoyl aspartate is cyclised. **Dihydo orotic acid** is produced. The enzyme is dihydro orotase (DHOase) (step 3, Fig. 39.18).

Step 4: Oxidation

Hydrogen atoms are removed from C5 and C6 positions, so that **orotic acid** is produced (step 4, Fig. 39.18). Enzyme is dihydro orotate dehydrogenase (DHODH). It requires NAD as co-enzyme.

Step 5: Formation of OMP

Ribose-5-phosphate is added to orotic acid, so as to produce orotidylic acid or orotidine monophosphate (**OMP**). PRPP is the donor of ribose-5-P. The enzyme is orotate phosphoribosyl transferase (OPRTase) (step 5, Fig. 39.18).

Step 6: Decarboxylation

The C7 of OMP is removed as carbon dioxide, so that uridine monophosphate (**UMP**) is produced (step 6, Fig. 39.18). This

is the **first pyrimidine that is synthesized**. The enzyme is OMP-decarboxylase (OMPDC). 6-aza-uridine inhibits this step, and so used as an anticancer drug.

Step 7, Synthesis of Triphosphates

UMP is phosphorylated to form **UDP** (uridine diphosphate) with the help of ATP (step 7, Fig. 39.18). The enzyme is nucleoside monophosphate kinase (UMP kinase). Next, the UDP is phosphorylated to **UTP** (uridine triphosphate) with the help of ATP (step 8, Fig. 39.18). The enzyme is nucleoside diphosphate kinase.

Step 8, Formation of CTP

UTP is converted to **CTP** by adding an amino group from glutamine catalysed by CTP synthetase. It needs ATP (step 9, Fig. 39.18).

Regulation of Pyrimidine Synthesis

i. In eukaryotes the first 3 enzymes, viz, CPS II, ATC and DHOase are present as a **multi-enzyme complex** and referred to as **'CAD'**, taking the first letters of the 3 enzymes.

ii. The last 2 enzymes, OPRTase and OMP decarboxylase are also present as a single functional complex. Because of this clustering of enzymes, the synthesis is well co-ordinated. Both complexes are cytosolic.

iii. The remaining enzyme, dihydro orotate dehydrogenase (step 4) is mitochondrial.

iv. The major regulatory step in prokaryotes is the reaction catalysed by aspartate trans-carbamoylase (ATC) which is allosterically inhibited by CTP.

v. In mammalian cells the regulation occurs at the level of **CPS II** (enzyme 1) which is inhibited by UTP are activated by PRPP. Aspartate trans carbamoylase (enzyme 2) is inhibited by CTP, and activated by ATP.

vi. Further, OMP decarboxylase is inhibited by UMP. The requirement of ATP for CTP formation and the stimulatory effect of GTP on CTP synthetase ensure a balanced synthesis of purine and pyrimidine nucleotides.

vii. Pyrimidines can also be **salvaged** like the purines, using PRPP and phosphoribosyl transferase and nucleoside phosphorylase.

viii. Both gene expression and enzyme activity are regulated. The first 3 and last 2 enzymes are regulated by repression / depression.

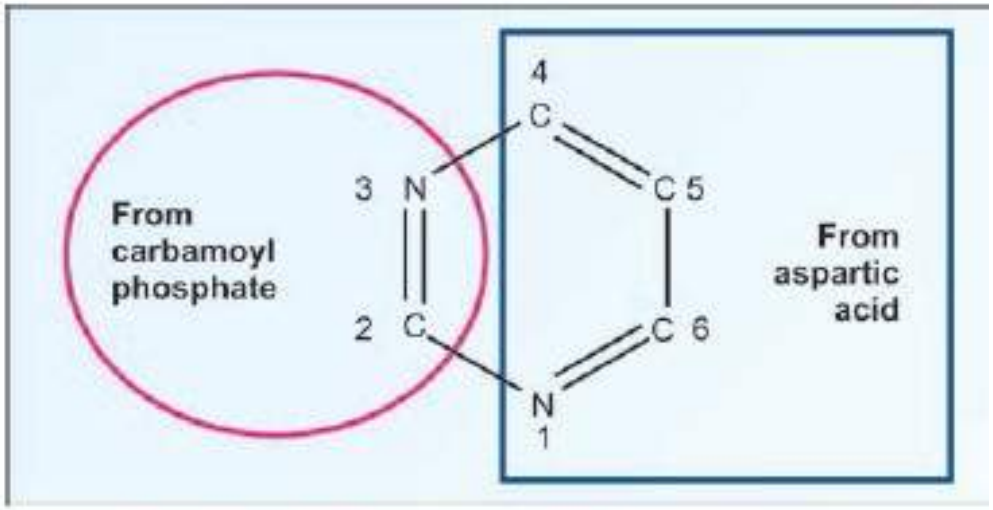
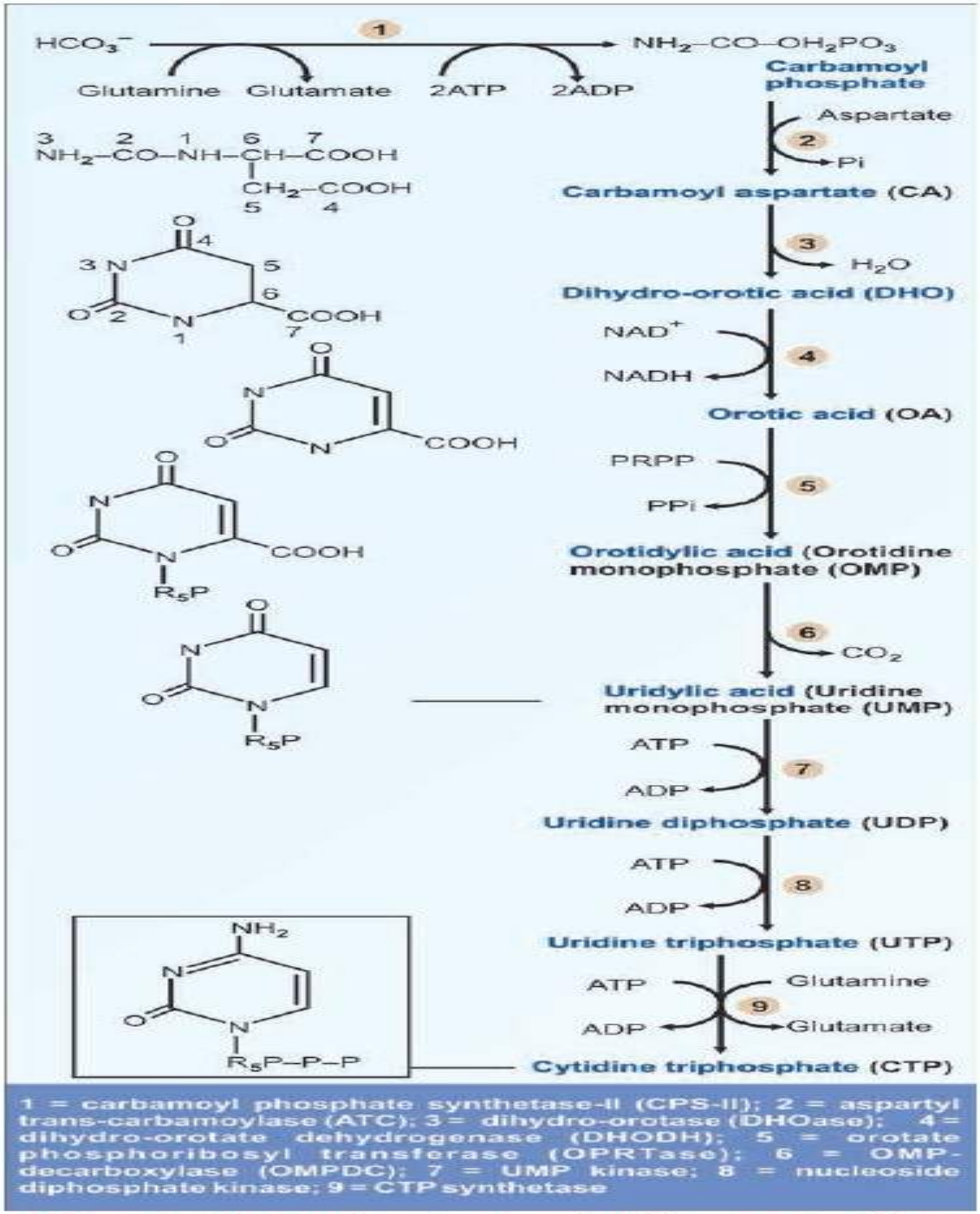


Fig. 39.17. Sources of C and N atoms of pyrimidine



1 = carbamoyl phosphate synthetase-II (CPS-II); 2 = aspartyl trans-carbamoylase (ATC); 3 = dihydro-orotase (DHOase); 4 = dihydro-orotate dehydrogenase (DHODH); 5 = orotate phosphoribosyl transferase (OPRTase); 6 = OMP-decarboxylase (OMPDC); 7 = UMP kinase; 8 = nucleoside diphosphate kinase; 9 = CTP synthetase

Fig. 39.18. Synthesis of pyrimidine nucleotides

Disorders of Pyrimidine Metabolism

Orotic Aciduria

i. The condition results from absence of either or both of the enzymes, **OPRTase and OMP decarboxylase**. It is an autosomal recessive disease.

ii. There is retarded growth and megaloblastic **anemia**. The rapidly growing cells are more affected and hence the anemia. Crystals are excreted in urine which may cause urinary tract obstruction. Due to lack of feedback inhibition orotic acid production is excessive.

iii. The condition can be successfully treated by feeding cytidine or uridine. They may be converted to UTP which can act as feedback inhibitor.

iv. Orotic aciduria may also occur in ornithine transcarbamoylase deficiency (**urea cycle** enzyme) as carbamoyl phosphate accumulates due to defective conversion to citrulline.

v. **Allopurinol** competes with orotic acid for the enzyme orotate phosphoribosyl transferase (enzyme no.5 in Fig. 39.18.), leading to orotic aciduria and orotidinuria

Deoxy Ribonucleotide Formation

Deoxy ribonucleotides (both purines and pyrimidine series) are formed by the reduction at the 2' carbon of the corresponding nucleoside diphosphates (**NDP to dNDP**). The enzyme is **ribonucleotide reductase complex**, which contains non-heme iron. It requires NADPH, **Thioredoxin** and thioredoxin reductase (Fig. 39.19). Thus, UDP is first converted to dUDP and then to dUTP.

Combined Regulation of Purine and Pyrimidine Synthesis

Purine and pyrimidines are synthesized in equimolecular quantities. This suggests co-ordinated control of their biosynthesis. PRPP is the precursor of both purines and pyrimidines; the PRPP synthase is inhibited by both purine and pyrimidine nucleotides. Both series of reactions are closely regulated by allosteric effectors. Binding of a specific NTP to the substrate-specificity site will have positive effect on the reduction of other NTPs (Table 39.6). Since ATP is required for the reduction of both CDP and UDP, the purine to pyrimidine balance is always maintained.

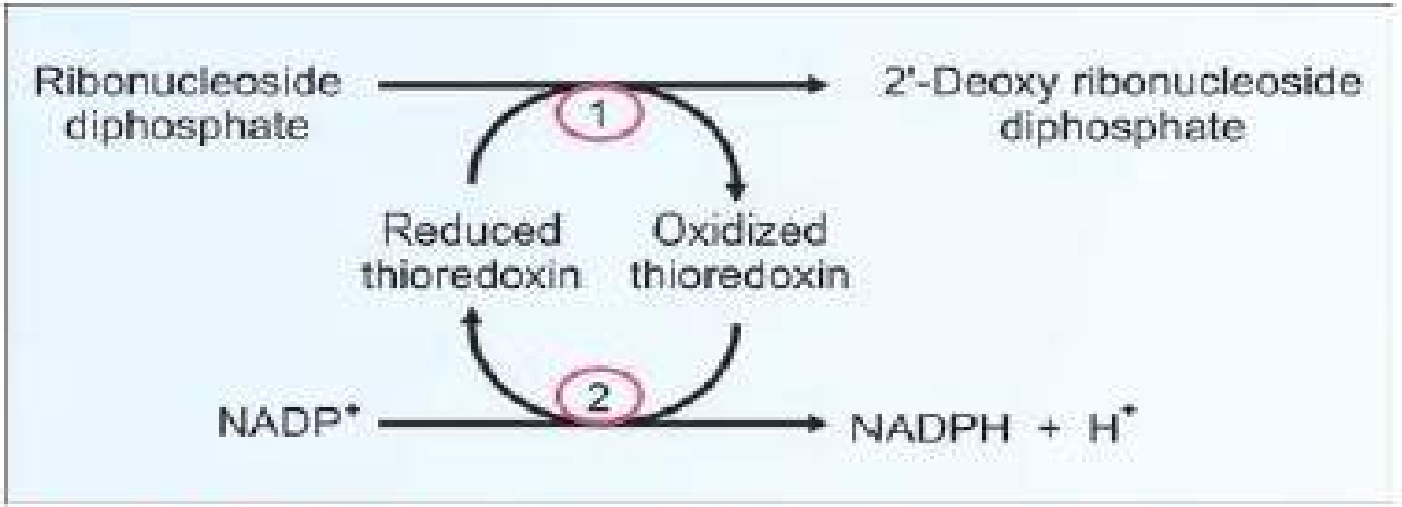


Fig. 39.19. Formation of 2'-deoxyribonucleoside diphosphates. 1 = Ribonucleotide reductase; 2 = Thioredoxin reductase

Synthesis of Deoxythymine Nucleotides

The thymine nucleotide is formed by **thymidylate synthase** by methylation of dUMP. The methyl group is donated by N^5,N^{10}-methylene-THFA. Later, THFA is regenerated by di hydro folate reductase, using NADPH as the reductant (Fig. 39.20). **Methotrexate** inhibits the enzyme DHF-reductase. So dTMP synthesis is inhibited, in turn DNA synthesis is inhibited.

Anticancer Agents Acting on Pyrimidines

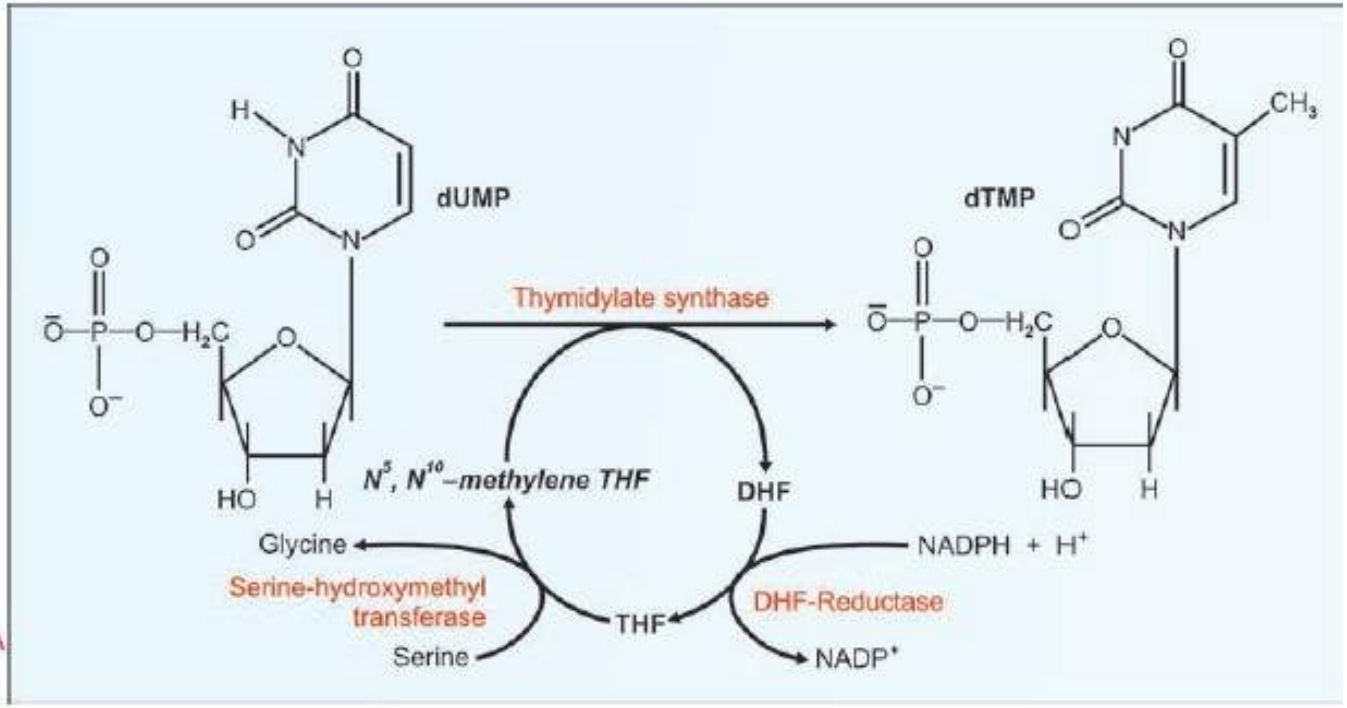
Methotrexate inhibits dihydrofolate reductase and thereby reduces the regeneration of THFA; it is a powerful anticancer agent (Fig. 39.20).

5-fluoro-uracil, 5-iodo uracil, 3-deoxy uridine, 6-aza uridine, 6-aza cytidine and 5-iodo-2-deoxy-uridine are anticancer drugs, which competitively inhibit thymidylate synthase. Cytosine arabinoside where ribose is replaced by arabinose is another anticancer agent.

Fig. 39.20.
Production of dTMP from dUMP, by the enzyme thymidylate synthase.

The reaction needs 1 carbon units, and folic acid. DHF = dihydro folic acid. THF = tetra-hydro folic acid.

Methotrexate inhibits the enzyme DHF-reductase. So dTMP synthesis is inhibited, in turn DNA synthesis is inhibited



Degradation of Pyrimidine Nucleotides

Uracil and thymine are degraded by analogous reactions. The phosphate is removed from nucleotide to form corresponding nucleoside. In the next step, free base is released. The ring is opened. Finally, **beta-amino isobutyric acid** is excreted in urine. This is the end product of pyrimidines. Other products are carbon dioxide and ammonia (Fig. 39.21). Pseudouridine is not metabolized further, and is excreted as such in urine.

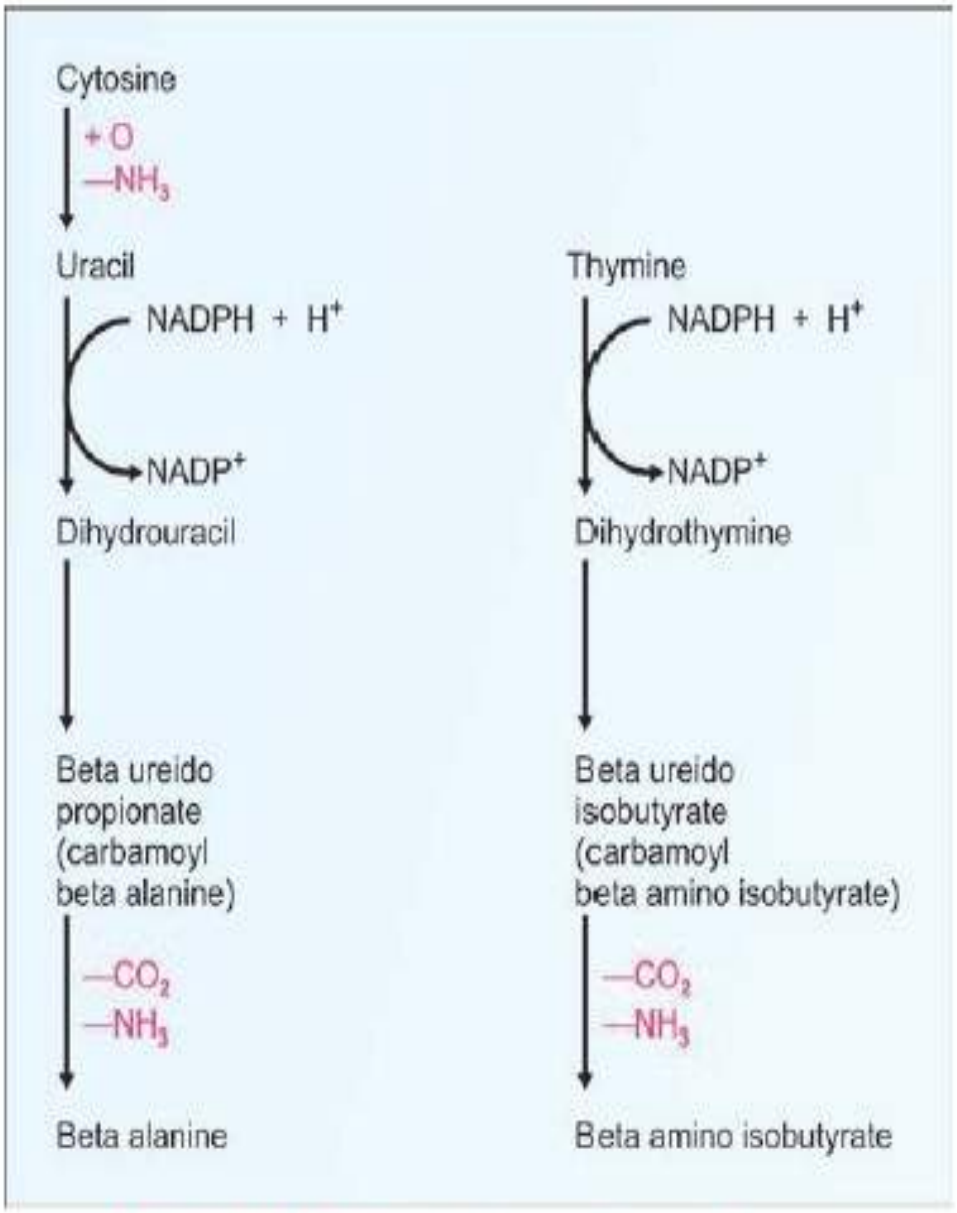


Fig. 39.21. Catabolism of pyrimidines

Table 39.6. Regulation of deoxyribonucleotide formation

CDP	→ d-CDP	ATP	dATP, dGTP, dTTP
UDP	→ d-UDP	ATP	dATP, dGTP
ADP	→ d-ADP	dGTP	dATP, ATP
GDP	→ d-GDP	dTTP	dATP

Reading

- Chatterjea
- Lippincott